COURSE: MACHINE LEARNING (LAB)

ML PROJECT PROPOSAL

MEMBERS:

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SECTION: **5A1**

DEGREE: **BS-AI**

<u>Dataset Selection:</u> <u>Multiple Disease Data</u>

<u>Target Variable Selection:</u> <u>Asthma</u>

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1. Problem Statement: Overcoming Asthma

Asthma is a chronic respiratory condition that millions of people all over the world (primarily children but also adults) suffer from, and which makes living more complicated as well as costing billions for health sector. The objective of this project is to build a machine learning model that can predict the probability of having asthma from input health indicators, lifestyle characteristics and demographic features. If we identify the high risk individuals that has propensity towards asthma at early stage, intervention in form of preventive measures and timely care by practitioners can control the disease and lessen its effect.

2. Brief Idea

The goal of the suggested machine learning model will be to categorize individuals according to their probabilities of suffering from asthma, using a number of features in terms of physical health, life style characteristics and other demographic trends. This project will utilize the predictive nature of machine learning to establish a robust device that could eventually aid health professionals in the early diagnosis and treatment of asthma.

3. Dataset Selection

In this project, we have chosen a dataset which contains health and lifestyle information for **59,068** individuals — allowing detailed exploration of the potential contribution of co-morbidities to asthma. In this case we will be treating the "Asthma" column as our label aka target variable to predict whether or not an individual has been diagnosed with asthma. The final dataset includes 16 features to be used for prediction, since the "KidneyDisease" and "SkinCancer" are irrelevant features; hence these feature had been removed from dataset.

Data Size (Dimensions): 59,068 samples (rows) x 16 features (columns)

4. Dataset Introduction

```
import pandas as pd

df = pd.read_csv("Multiple Disease Data.csv")
df.drop(columns = ['KidneyDisease', 'SkinCancer'], inplace = True)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 59068 entries, 0 to 59067
Data columns (total 16 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   HeartDisease      59068 non-null  object
 1   BMI               59068 non-null  float64
 2   Smoking           59068 non-null  object
 3   AlcoholDrinking   59068 non-null  object
 4   Stroke            59068 non-null  object
 5   PhysicalHealth    59068 non-null  int64
 6   MentalHealth      59040 non-null  float64
 7   DiffWalking       59068 non-null  object
 8   Sex               59068 non-null  object
 9   AgeCategory       59068 non-null  object
 10  Race              59068 non-null  object
 11  Diabetic          59068 non-null  object
 12  PhysicalActivity  59068 non-null  object
 13  GenHealth         59059 non-null  object
 14  SleepTime         59068 non-null  int64
 15  Asthma            59068 non-null  object
dtypes: float64(2), int64(2), object(12)
memory usage: 7.2+ MB
```

The dataset includes a number of categorical and numeric features regarding health, lifestyle, and demographic characteristics that could potentially be associated with asthma risk. Key features include:

a. **BMI (Body Mass Index):** A possible manifestation of respiratory health

b. **PhysicalHealth, MentalHealth:** Subjective physical and mental health status measurements that may or may not be relevant to the start of asthma.

c. **AlcoholDrinking, Smoking:** Variables that may influence some aspects of respiratory health.

d. **AgeCategory, Sex, Race:** Demographic information that may provide insight on asthma trends among various population groups.

5. Classifier Selection

We will test the following classifiers to classify whether a person is most likely to have asthma:

a. **Logistic Regression:** Usually employed for binary classification problems and easy to interpret.

b. **Random Forest:** A strong ensemble model that works with a large variety of features and interactions.

c. **Support Vector Machine (SVM):** The SVM works great on high-dimension data and complex classification tasks.

In this study, we evaluate each classifier using accuracy, precision, recall and F1-score in order to identify the best performing model for this dataset.

6. What to Achieve

The project has the following key objectives:

a. For constructing a model which will be able to predict the risk for asthma given various features of an input

b. To find the important predictors of asthma within the dataset to shed some light into prevention strategies.

c. For development of a tool that health care providers can use to do screening for asthma risk prior to diagnosis.

7. Methodology

a. **Data Preprocessing**: This involves filling in any missing values, coding categorical variables, and scaling numerical features to ensure that you have a consistent representation of your data.

b. **Exploratory Data Analysis (EDA):** We will perform EDA to look for distributions, correlations and skews in terms of asthma features.

c. **Model Training & Evaluation:** A number of classifiers trained on the data and evaluated using cross-validation, while accuracy, precision, recall as well as F1-score to select the best model.

d. **Strength of Features:** Here, we will also visualize the strength features to know about which factor influence asthma predictions most.

e. **Model Optimization:** Hyper parameter tuning can help improve both model performance and generalizability.